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Stability of networked systems: a multi-scale approach using contraction

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***Abstract*— This paper is concerned with the stability of networked control systems. Using contraction theory, a result is established on the network structure and the properties of the individual component subsystems and their couplings to ensure the overall contractivity of the entire network. Specifically, it is shown that a contraction property on a reduced-order matrix that quantifies the interconnection structure, coupled with contractivity/expansion estimates on the individual component subsystems, suffices to ensure that nearby trajectories of the overall system converge towards each other. The paper also describes a relative contraction result of interest in synchronization problems.**

I. Introduction

It is often useful to break down the analysis and design of large-scale systems into two independent steps:

(a) At a "global" level, properties of a network or interconnection graph are imposed so as to guarantee a desired behavior for the full interconnected system. In this analysis, subsystems may be characterized as "black boxes" with assumed input-output characteristics, but detailed knowledge of their internal structure is not required.

(b) At a "local" level of analysis, one imposes constraints on the structure and behavior of individual subsystems (components), so as to fit the requirements of the global approach. These requirements are verified independently of the overall network structure. This "multi-scale" or hierarchical methodology is robust in so far as a large degree of uncertainty can be tolerated in the components, only constrained by meeting appropriate behavioral requirements.

There are many examples of such approaches in control theory, including among others (1) the use of small-gain theorems to guarantee stability of a negative feedback loop provided that the components are individually stable (qualitative property of components) and the overall loop H_∞ gain is less than one, as well as nonlinear generalizations based on input to state stability [1], [2]; (2) input/output monotone systems theory [3], [4], [5], in which input-output characteristics are the only required "quantitative" data; (3) the use of passivity-based tools [6], [7], [8].

In this work, we present yet another example of the general principle, this time in the framework of contractive systems discussed in [9]. We show that contractivity of the overall system can be guaranteed if the matrix measures of Jacobians of individual components are upper bounded and the measure of a reduced-order matrix associated to the interconnection is negative. Note that the construction of such reduced-order matrix uses only norm and matrix-measure estimates, but no precise knowledge, of components. No assumptions are made on the networks. Directed networks, self-loops, and multiple regulatory interactions are allowed.

Besides the above-discussed advantages of hierarchical verification and design, there are other potential computational advantages of such an approach. Indeed, computing matrix measures for nonlinear systems involves maximizations of measures of Jacobians over the entire state-space, a computationally intensive task. On the other hand, our approach permits one to perform this maximization over each individual subsystem and coupling function (and separately verify contractivity of the reduced-order matrix). Each of the subsystems has lower dimension than the overall system, lowering the computational load. Of course, doing so may provide estimates that are too conservative, but the method is "best possible", in the sense that for special interconnection structures the same bounds are obtained (this happens, for example, when the systems happen to be decoupled).

II. A globalization result with matrix measures

We recall (see for instance [10]) that, given a vector norm on Euclidean space ($|\bullet|$), with its induced matrix norm $\|A\|$, the associated *matrix measure* μ is defined as the directional derivative of the matrix norm, that is,

$$\mu(A) := \lim_{h \searrow 0} \frac{1}{h} \left(\|I + hA\| - 1\right).$$

For example, if $|\bullet|$ is the standard Euclidean 2-norm, then $\mu(A)$ is the maximum eigenvalue of the symmetric part of A. Matrix measures are also known as "*logarithmic norms*", a concept independently introduced by Germund Dahlquist and Sergei Lozinskii in 1959, [11], [12]. The limit is known to exist, and the convergence is monotonic, see [13], [11].

We assume given: (i) k spaces $\mathbb{R}^{n_i}$ endowed respectively with "local" norms $|\xi_i|_{\mathrm{L,i}}$, $i = 1, \ldots, k$, and (ii) an "interconnection" or "structure" norm $|x|_{\mathrm{S}}$ on $\mathbb{R}^k$. The structure norm is assumed to be *monotone*, meaning that, for any two vectors $x, y \in \mathbb{R}^k$,

$$0 \le x \le y \quad \Rightarrow \quad |x|_{\mathrm{S}} \le |y|_{\mathrm{S}}$$

where an inequality such as "$x \le y$" between vectors is understood coordinate-wise, that is, $x_i \le y_i$ for all indices i. All the usual L^p norms, with $1 \le p \le \infty$ are monotone.

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We let

$$N \;:=\; n_1 + n_2 + \ldots + n_k$$

and introduce a "global" norm on $\mathbb{R}^N$ as follows. Given any vector $\xi = (\xi_1, \ldots, \xi_k)^{\mathrm{T}} \in \mathbb{R}^N$, with $\xi_i \in \mathbb{R}^{n_i}$, $i = 1, \ldots, k$,

$$|\xi|_{\mathrm{G}} \;:=\; \left|\left(|\xi_1|_{\mathrm{L},1}, \ldots, |\xi_k|_{\mathrm{L},k}\right)^{\mathrm{T}}\right|_{\mathrm{S}} .$$

That is, the global norm is obtained by first computing the local norms $x_i = |\xi_i|_{\mathrm{L,i}}$, and then evaluating the structure norm of the resulting vector x with components x_i. Using that the structure norm is assumed to be monotone, it is easy to show that this is indeed a norm,

For example, if all the local norms as well as the structure norms are L^p norms, with the same p, then the global norm is again the same L^p norm (on a larger space). However, more generally one may mix different norms.

Given norms $|x|_1$ and $|y|_2$ in $\mathbb{R}^q$ and $\mathbb{R}^p$ respectively, we may consider the usual induced operator norm on matrices $A \in \mathbb{R}^{p\times q}$, $\|A\|_{12} := \sup_{|x|_1=1} |Ax|_2$. In particular, for the special cases when $p = n_i$ and $q = n_j$ and the above norms, we denote this norm as $\|A\|_{\mathrm{L,i,j}}$:

$$\|A\|_{\mathrm{L,i,j}} \;:=\; \sup_{|x|_{\mathrm{L,i}}=1} |Ax|_{\mathrm{L,j}} .$$

When $p = q = k$ we write:

$$\|A\|_{\mathrm{S}} \;:=\; \sup_{|x|_{\mathrm{S}}=1} |Ax|_{\mathrm{S}} ,$$

and when $p = q = N$:

$$\|A\|_{\mathrm{G}} \;:=\; \sup_{|x|_{\mathrm{G}}=1} |Ax|_{\mathrm{G}} .$$

We use the notations $\mu_{\mathrm{L,i}}[\cdot]$, $\mu_{\mathrm{S}}[\cdot]$, and $\mu_{\mathrm{G}}[\cdot]$ for the matrix measures (logarithmic norms) associated to $\|\cdot\|_{\mathrm{L,i,i}}$, $\|\cdot\|_{\mathrm{S}}$, and $\|\cdot\|_{\mathrm{G}}$ respectively.

Given any "global" matrix $A_{\mathrm{G}} \in \mathbb{R}^{N\times N}$, we define its associated "structure" matrix $A_{\mathrm{S}} \in \mathbb{R}^{k\times k}$ as follows. We start by partitioning A_{G} in the form:

$$A_{\mathrm{G}} \;=\; \begin{pmatrix} A_{11} & A_{12} & \ldots & A_{1k} \\ A_{21} & A_{22} & \ldots & A_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ A_{k1} & A_{k2} & \ldots & A_{kk} \end{pmatrix}$$

and for each $i = 1, \ldots, k$, we define the following numbers:

$$\widetilde{A}_{ii} \;:=\; \mu_{\mathrm{L,i}}[A_{ii}]$$

and for each $i, j \in \{1, \ldots, k\}$ with $i \neq j$, we let:

$$\widetilde{A}_{ij} \;:=\; \|A_{ij}\|_{\mathrm{L,i,j}} .$$

Finally, we define:

$$A_{\mathrm{S}} \;:=\; \begin{pmatrix} \widetilde{A}_{11} & \widetilde{A}_{12} & \ldots & \widetilde{A}_{1k} \\ \widetilde{A}_{21} & \widetilde{A}_{22} & \ldots & \widetilde{A}_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ \widetilde{A}_{k1} & \widetilde{A}_{k2} & \ldots & \widetilde{A}_{kk} \end{pmatrix} .$$

(The proofs will actually show a little more, namely that upper bounds on the $\widetilde{A}_{ij}$ could be used, instead.)

Our main result is as follows:

Theorem 1: For every set of "local" norms on $\mathbb{R}^{n_i}$, every "structure" norm on $\mathbb{R}^k$, and every matrix $A_{\mathrm{G}} \in \mathbb{R}^{N\times N}$,

$$\mu_{\mathrm{G}}[A_{\mathrm{G}}] \;\leq\; \mu_{\mathrm{S}}[A_{\mathrm{S}}] .$$

This theorem follows from:

Lemma 1: For every set of "local" norms on $\mathbb{R}^{n_i}$, every "structure" norm on $\mathbb{R}^k$, and every matrix $A_{\mathrm{G}} \in \mathbb{R}^{N\times N}$:

$$\|I + hA_{\mathrm{G}}\|_{\mathrm{G}} \;\leq\; \|I + hA_{\mathrm{S}}\|_{\mathrm{S}} + g(h) \qquad \text{for all } h > 0 , \tag{1}$$

where $g : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ is such that $g(h) = o(h)$ as $h \searrow 0$.

To see how Theorem 1 follows from Lemma, we recall that $\mu_{\mathrm{G}}[A_{\mathrm{G}}] = \lim_{h\searrow 0} \frac{1}{h}\left(\|I + hA_{\mathrm{G}}\|_{\mathrm{G}} - 1\right)$, and similarly for $\mu_{\mathrm{S}}[A_{\mathrm{S}}]$. Subtracting 1 from both sides in (1), dividing by h, and taking the limit as $h \searrow 0$, the Theorem results.

We now prove the Lemma.

Pick any vector $\xi \in \mathbb{R}^N$. We will show that

$$|(I + hA_{\mathrm{G}})\xi|_{\mathrm{G}} \;\leq\; \left[\|I + hA_{\mathrm{S}}\|_{\mathrm{S}} + g(h)\right] |\xi|_{\mathrm{G}} \tag{2}$$

for some function g as above. Since this holds in particular for all ξ with $|\xi|_{\mathrm{G}} = 1$, the Lemma will follow.

We need the following observation. Since, for any norm $|\cdot|$ and induced matrix norm $\|\cdot\|$, by definition the matrix measure of a matrix B is $\mu(B) = f'(0)$, where $f(h) = \|I + hB\|$, there is a function $g(h) = o(h)$ such that $\|I + hB\| = 1 + h\mu(B) + g(h)$. In particular, there are such functions $g_i(h)$ associated to the "local" norms $|\cdot|_{\mathrm{L,i}}$. We let $g(h) := \max\{g_1(h), \ldots, g_k(h)\}$. Thus,

$$\|1 + hA_{ii}\|_{\mathrm{L,i}} \;\leq\; 1 + h\mu_{\mathrm{L,i}}[A_{ii}] + g(h),\; i = 1, \ldots, k . \tag{3}$$

Note that $g(h) = o(h)$.

We start the proof of the Lemma by writing in block form $\xi = (\xi_1, \ldots, \xi_k)^{\mathrm{T}}$, with $\xi_i \in \mathbb{R}^{n_i}$, $i = 1, \ldots, k$, and denote

$$\eta \;:=\; (I + hA_{\mathrm{G}})\,\xi .$$

In block form, $\eta = (\eta_1, \ldots, \eta_k)^{\mathrm{T}}$, with $\eta_i \in \mathbb{R}^{n_i}$, $i = 1, \ldots, k$, where

$$\eta_i \;=\; (1 + hA_{ii})\xi_i \;+\; \sum_{j\neq i} hA_{ij}\xi_j .$$

By definition of the global norm,

$$|\eta|_{\mathrm{G}} = |x|_{\mathrm{S}}$$

where we denote $x_i := |\eta_i|_{\mathrm{L,i}}$ for each $i = 1, \ldots, k$ and $x = (x_1, \ldots, x_k)^{\mathrm{T}}$. Similarly,

$$|\xi|_{\mathrm{G}} = |y|_{\mathrm{S}}$$

where we denote $y_i := |\xi_i|_{\mathrm{L,i}}$ for each $i = 1, \ldots, k$ and $y = (y_1, \ldots, y_k)^{\mathrm{T}}$.

Using the triangle inequality, we have that, for every $i = 1, \ldots, k$:

$$\begin{aligned}
x_i &\leq |(1 + hA_{ii})\xi_i|_{\mathrm{L,i}} + \sum_{j \neq i} |hA_{ij}\xi_j|_{\mathrm{L,i}} \\
&\leq \|1 + hA_{ii}\|_{\mathrm{L,i}} |\xi_i|_{\mathrm{L,i}} + \sum_{j \neq i} h \|A_{ij}\|_{\mathrm{L,i,j}} |\xi_j|_{\mathrm{L,j}} \\
&\leq \left[1 + h\mu_{\mathrm{L,i}}[A_{ii}]\right] |\xi_i|_{\mathrm{L,i}} + \sum_{j \neq i} h \|A_{ij}\|_{\mathrm{L,i,j}} |\xi_j|_{\mathrm{L,j}} \\
&\qquad + g(h) |\xi_i|_{\mathrm{L,i}} \\
&= z_i := \left[1 + h\widetilde{A}_{ii}\right] y_i + \sum_{j \neq i} h\widetilde{A}_{ij} y_j + g(h) y_i ,
\end{aligned}$$

where we used the estimate (3). In terms of the following vector $z \in \mathbb{R}^k$:

$$z := (I + hA_{\mathrm{S}})\, y + g(h)\, y ,$$

we can summarize the above inequality as "$x \leq z$" (in the coefficient-wise order), and hence, using monotonicity of the "structure" norm, we know that $|x|_{\mathrm{S}} \leq |z|_{\mathrm{S}}$. By the triangle inequality,

$$|x|_{\mathrm{S}} \leq \|I + hA_{\mathrm{S}}\|_{\mathrm{S}} |y|_{\mathrm{S}} + g(h)\, |y|_{\mathrm{S}} ,$$

Recalling that $|x|_{\mathrm{S}} = |\eta|_{\mathrm{G}} = |(I + hA)\,\xi|_{\mathrm{G}}$ and $|y|_{\mathrm{S}} = |\xi|_{\mathrm{G}}$, we have that

$$|(I + hA_{\mathrm{G}})\,\xi|_{\mathrm{G}} \leq \left[\|I + hA_{\mathrm{S}}\|_{\mathrm{S}} + g(h)\right] |\xi|_{\mathrm{G}}$$

which is (2) and hence the Lemma is proved. ∎

III. Application to structured systems

We consider systems of ordinary differential equations, generally time-dependent:

$$\dot{x} = f(t, x) \tag{4}$$

defined for $t \in [0, \infty)$ and $x \in C \subseteq \mathbb{R}^n$. We assume that $f(t, x)$ is differentiable on x, and that $f(t, x)$, as well as the Jacobian of f with respect to x, denoted as $J(t, x) = \frac{\partial f}{\partial x}(t, x)$, are both continuous in (t, x). We denote by $\varphi(t, s, \xi)$ the value of the solution $x(t)$ at time t of the differential equation (4) with initial value $x(s) = \xi$; we will assume that $\varphi(t, s, \xi)$ is defined and belongs to C for all $t \geq s$.

We say that system (4) is *(infinitesimally) contracting* if there exists some norm in $\mathbb{R}^n$, with associated matrix measure μ, such that, for some constant $c > 0$, the *contraction rate*:

$$\mu\left(J\left(x, t\right)\right) \leq -c, \quad \forall x \in C, \quad \forall t \geq 0\,. \tag{5}$$

The key theoretical result about contracting systems links infinitesimal contraction to global contractivity, and is stated below. This result and its proof can be found in [9].

Theorem 2: Suppose that C is a convex subset of $\mathbb{R}^n$ and that $f(t, x)$ is (infinitesimally) contracting with contraction rate c. Then, for every two solutions $x(t) = \varphi(t, 0, \xi)$ and $z(t) = \varphi(t, 0, \zeta)$ of (4), it holds that:

$$|x(t) - z(t)| \leq e^{-ct} |\xi - \zeta| , \qquad \forall\, t \geq 0\,. \tag{6}$$

In [14], an alternative proof of Theorem 2 is given, and other references to the history of this notion are presented (see also [15]).

The above general results on contraction can be immediately applied to the study of global stability of interconnected systems.

For example, if the interconnection structure is a graph with pure diffusion among components, then it is convenient to employ Euclidean norm for the structure norm. The structure measure will evaluate to zero, which implies that the interconnection structure does not destroy contraction.

More generally, one would like to study synchronization. In this case, contraction to the set of states with equal coordinates will be the goal, and the (negativity of the) second eigenvalue of the Laplacian matrix will determine the structure measure.

IV. Relative contractions

We now revisit the question of contractions relative to a subspace M originally posed in [16].

Let us pick any orthogonal matrix Q ($QQ^{\mathrm{T}} = Q^{\mathrm{T}}Q = I$), and partition Q as:

$$Q = (VW) \in \mathbb{R}^{n \times n}$$

$V = (v_1, \ldots, v_q) \in \mathbb{R}^{n \times q}$ and $W = (w_1, \ldots, w_m) \in \mathbb{R}^{n \times m}$. Since Q is orthogonal, $VV^{\mathrm{T}} + WW^{\mathrm{T}} = I$, $W^{\mathrm{T}}W = I$, $V^{\mathrm{T}}V = I$, $W^{\mathrm{T}}V = 0$, and $V^{\mathrm{T}}W = 0$ (zero and identity matrices of appropriate sizes). Let M be the linear subspace of $\mathbb{R}^n$ spanned by the columns of $W = (w_1, \ldots, w_m) \in \mathbb{R}^{n \times m}$. Equivalently, the columns of V constitute an orthogonal basis of $M^{\perp}$, i.e. $M = \ker V^{\mathrm{T}}$,. In applications to interconnected systems, M will typically consist of those states of the composite system in which all the components are the same. We consider the following conditions on C:

$$V\mathbb{R}^q + WW^{\mathrm{T}}C \subseteq C \tag{7}$$

(this condition is satisfied in the special case that $C = \mathbb{R}^n$) and on f:

$$V^{\mathrm{T}}f(t, WW^{\mathrm{T}}C) \subseteq C\,. \tag{8}$$

Since M is the range of W and the kernel of V^{T}, condition (8) is satisfied if we know that M is a forward-invariant set, $f(t, M) \subseteq M$.

We say that system (4) is *infinitesimally contracting to* M if there exists some norm in $\mathbb{R}^n$, with associated matrix measure μ, such that, for some constant $c > 0$, the *contraction rate*:

$$\mu\left(V^{\mathrm{T}}J\left(x,t\right)V\right) \leq -c, \quad \forall x \in C, \quad \forall t \geq 0 \tag{9}$$

The main result is now:

Theorem 3: Suppose that C, M, V satisfy conditions (7) and (8), and that $f(t, x)$ is infinitesimally contracting to M with contraction rate c. Consider any solution $x(t) = \varphi(t, 0, \xi)$ of (4). Then

$$\left|V^{\mathrm{T}}x(t)\right| \leq e^{-ct}\left|V\xi\right|, \qquad \forall\, t \geq 0\,. \tag{10}$$

Since $V^{\mathrm{T}}x(t) \to 0$ as $t \to \infty$ means that $x(t) \to \ker V^{\mathrm{T}} = M$, this result implies that every trajectory of (4) exponentially approaches M.

Proof: Fix the particular solution $\bar{x}(t) = \varphi(t, 0, \xi)$ of (4) of interest. Let $\eta := V^{\mathrm{T}}\bar{x}(0) \in \mathbb{R}^q$. We consider the following system of differential equations in $\mathbb{R}^q$:

$$\dot{y} = g(t, y) := V^{\mathrm{T}}f(t, Vy + WW^{\mathrm{T}}\bar{x}(t))\,. \tag{11}$$

For each $r \in [0, 1]$, let

$$\psi\,(t, r) := \text{ solution of (11) with } \psi\,(0, r) = r\,\eta\,.$$

The function $\psi\,(t, r)$ is continuously differentiable jointly on (t, r).

Observe that $\bar{x}(t) = (VV^{\mathrm{T}} + WW^{\mathrm{T}})\bar{x}(t) = V\bar{y}(t) + WW^{\mathrm{T}}\bar{x}(t)$, where $\bar{y}(t) := V^{\mathrm{T}}\bar{x}(t)$. Since $\bar{x}$ satisfies $\dot{x} = f(t, x)$, it follows that $\dot{\bar{y}} = V^{\mathrm{T}}f(t, \bar{x}) = g(t, \bar{y}(t))$. This means that $\bar{y}(t)$ is a solution of (11). Since $\bar{y}(0) = V^{\mathrm{T}}\bar{x}(0) = \eta$, it follows by uniqueness of solutions that $\psi\,(t, 1) = \bar{y}(t)$ for all $t \geq 0$.

Similarly, $y \equiv 0$ is also a solution, because $V^{\mathrm{T}}f(t, WW^{\mathrm{T}}\bar{x}(t)) \equiv 0$ (by condition (8). Thus, $\psi\,(t, 0) = 0$ for all $t \geq 0$.

We define

$$w(t, r) := \frac{\partial\psi}{\partial r}(t, r).$$

Since $\psi(0, r) = r\eta$, we have that $w(0, r) = \eta$ for all r. Further,

$$\begin{aligned}\frac{\partial w}{\partial t}(t, r) &= \frac{\partial}{\partial t}\left(\frac{\partial\psi}{\partial r}\right) = \frac{\partial}{\partial r}\left(\frac{\partial\psi}{\partial t}\right)\\ &= \frac{\partial}{\partial r}\,g(t, \psi\,(t, r)) = A(t, r)\,w(t, r)\end{aligned}$$

where

$$A(t, r) = V^{\mathrm{T}}J(t, x(t, r))V$$

and where

$$x(t, r) := V\psi\,(t, r) + WW^{\mathrm{T}}\bar{x}(t) \in C$$

by property (7).

Coppel's inequality [17] yields:

$$|w(t, r)| \leq |w(0, r)|\,e^{\int_0^t \mu(A(\tau, r))d\tau} \leq |\eta|\,e^{-ct}\,, \tag{12}$$

for all $t \geq 0$ and all $r \in [0, 1]$. From

$$\bar{y}(t) - 0 = \psi\,(t, 1) - \psi\,(t, 0) = \int_0^1 w(t, r)\,dr$$

it now follows that

$$|\bar{y}(t)| \leq \int_0^1 |w(t, r)|\,dr \leq |\eta|\,e^{-ct}\,,$$

and the proof is complete because $\eta = V^{\mathrm{T}}\bar{x}(0) = V^{\mathrm{T}}\xi$ and $\bar{y}(t) = V^{\mathrm{T}}\bar{x}(t)$. ∎

V. Applications

We now sketch several ways in which the contraction results described so far can be applied to networked control systems.

We consider interconnected systems:

$$\dot{x}_i = \phi_i(t, x) := f_i(t, x_i) + \tilde{h}_i(t, x), \tag{13}$$

with $i = 1, \ldots, N$ and $x = [x_1^T\ x_2^T\ \ldots x_N^T]^T$.

In (13) the function f_i is the intrinsic dynamics of the i-th node, while the function $\tilde{h}_i$, termed as interaction or coupling function, describes the interaction of the i-th node with the other nodes composing the interconnected system. A particular choice for the functions $h_i(t, x)$ is:

$$\tilde{h}_i(t, x) = h_i(t, a_{i1}(t)x_1, a_{i2}(t)x_2, \ldots, a_{iN}(t)x_N),$$

where $A(t) := [a_{ij}(t)]$ is the $N \times N$ time-varying adjacency matrix, with $a_{ij}(t) : \mathbb{R}^+ \to [0, 1]$ being smooth functions. Notice that the above formalization allows us to consider within a unique framework directed and undirected networks with (smoothly) changing topology, self loops and multiple interactions. Of course, one could incorporate the time-varying coefficients $a_{ij}(t)$ directly into the functions h_i, redefining this function, so that there is no need for the "tilde" in Equation (13). However, we prefer to write things in this form in order to emphasize that the entries of $A(t)$ may be uncertain.

We denote with $J := [J_{ij}]$ the Jacobian of the interconnected system composed by the N subsystems in (13), i.e.

$$J_{ij} := \frac{\partial f_i}{\partial x_j} + \frac{\partial \tilde{h}_i}{\partial x_j}. \tag{14}$$

In what follows we assume that there are numbers $c_i \in \mathbb{R}$ and non-negative numbers $m_{ij} \geq 0$ such that:

- $\mu_{L,i}(J_{ii}(t,x)) \leq c_i$, for all $x \in S$ and for all $t \in \mathbb{R}^+$;
- $\|J_{ij}(t,x)\|_{L,i,j} = \left\| \frac{\partial \tilde{h}_i}{\partial x_j} \right\| \leq m_{ij}$, $i \neq j$, for all $x \in S$ and for all $t \in \mathbb{R}^+$, where $\|\bullet\|$ is the operator norm on linear operators $\mathbb{R}^{n_j} \to \mathbb{R}^{n_i}$.

All norms in Euclidean space being equivalent, the finiteness of the upper bounds m_{ij} for one norm will imply the existence of such upper bounds for any other norm. However, the actual values are critical for the estimates.

Using Theorem 1, contractivity of the overall system can be established from the contractivity of the reduced-order matrix A_{S} obtained from the above norms; A_{S} can be constructed as shown in Section II with $A = J$ evaluated at all possible points in the state space and times.

Remarks

1. Note that, if for all $x \in S$, $t \in \mathbb{R}^+$, and $i = 1, \ldots, N$, we have:

$$\mu_{L,i}\left(\frac{\partial f_i}{\partial x_i}\right) \leq \alpha, \text{ and } \mu_{L,i}\left(\frac{\partial \tilde{h}_i}{\partial x_i}\right) \leq m_{ii},$$

then, in terms of the previous notation, we can take $c_i = \alpha + m_{ii}$. It follows that we can write $A_{\mathrm{S}} = \alpha I + M$, where $M = [m_{ij}]$ and I is the identity matrix of appropriate dimensions.

Hence, contraction of the entire system is guaranteed if

$$\mu_G(J) \leq \mu_S(A_{\mathrm{S}}) = \mu_S(\alpha I) + \mu_S(M) = \alpha + \mu_S(M) < 0\,. \tag{15}$$

2. When applied to networked systems, our results ensure that all nodes globally exponentially converge to a unique point on the synchronization manifold $x_1 = \ldots = x_N$ as in our case such a manifold is embedded into a contracting space. In this sense, the results described so far guarantee a stronger result than just convergence of all trajectories towards the synchronization manifold in state space. This is illustrated via numerical simulations in the next section.

VI. Numerical examples

We consider the problem of designing a strategy to guarantee contraction of a network consisting of Hopfield (HN for short) models [18]. We assume that the coupling functions are not necessarily diffusive, while the network topology is directed. We remark here that in our approach the individual subsystems are considered as black boxes, characterized by their contraction estimates. Hence, the same coupling functions derived below can be used to ensure stability of networks of different subsystems characterized by the same contraction estimates.

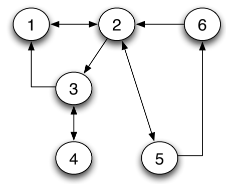

Fig. 1. Network used for the simulations of HN

TABLE I
Coupling function for network (16)

edge	$h_{ij}(x)$
$1 \leftrightarrow 2$, $6 \to 2$, $2 \to 3$, $5 \to 6$	$G \arctan(x)$
$3 \to 1$, $5 \leftrightarrow 2$	Kx
$3 \leftrightarrow 4$	$\frac{1-e^{-x}}{1+e^{-x}}$

Synchronization of HN: We start by considering the network topology of Figure 1. The model we consider is a network of HN [18] given by:

$$\dot{x}_i = -\frac{x_i}{R} + \sum_{j \in N_i} (h_{ij}(x_j) - h_{ij}(x_i)) + u(t), \tag{16}$$

where $i = 1, \ldots, 6$. Here, x_i denote the neural voltages, R is the resistance, h_{ij} are the coupling functions, N_i is the set of neighbours of node i and $u(t)$ represents some periodic external input acting on all nodes. According to Theorem 1, contraction is guaranteed if the measure of the reduced-order matrix A_S constructed as in Section II is uniformly negative.

In our case, the network has $f_i(x_i,t) = -\frac{x_i}{R} + u(t)$ and $\tilde{h}_i(x) := \sum_{j \in N_i} (h_{ij}(x_j) - h_{ij}(x_i))$.

Deriving the Jacobian of the interconnected system as in (14), we then have

$$J(x) = A_{\mathrm{S}} = -\frac{1}{R} I + M(x),$$

where I is the identity matrix of appropriate dimensions and $M(x) := \left\{ \frac{\partial h_{ij}}{\partial x_j} \right\}$. In order to guarantee contraction, we choose the coupling functions between nodes as those reported in Table I and use Theorem 1 to find simple conditions on the coupling gains G and K in order to ensure network stability. Using as structure matrix measure the one induced by the weighted vector-1 norm, we have:

$$\mu_{1,P}\left(J(x)\right) \leq -\frac{1}{R} + \mu_{1,P}\left(M(x)\right).$$

The simplest choice for the matrix measure $\mu_{1,P}$ is that induced by the weights $p_1 = \ldots = p_N$, obtaining:

$$\mu_{1,P}\left(M(x)\right) \leq \max\left\{-G + K, G\right\}.$$

Now, if we set $G < 1/R$ and $K - G < 1/R$, then the Jacobian measure is uniformly negative definite, making the network contracting. Since each node is forced by the periodic function $u(t)$, this implies that there exist a unique periodic orbit towards which all trajectories of (16) converge (see [14]). Furthermore, it is straightforward to check that the synchronization manifold defined as the subspace where $x_1 = \ldots = x_6$ is flow invariant for the network dynamics. Thus, we can conclude that contraction of (16) implies that all of its trajectories converge towards a unique solution which is embedded into the synchronization manifold. That is, at steady state network nodes are synchronized onto a periodic orbit having the same period as $u(t)$.

As shown in Figure 2, the numerical simulations confirm the theoretical predictions.

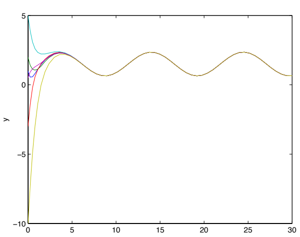

Fig. 2. Simulation of network (16) with $R = 1$, $G = K = 0.9$. Network state variables on the y-axis. Time on the x-axis.

VII. CONCLUSIONS

We presented sufficient conditions ensuring stability of networked systems, using contraction. The analysis is multi-scale, and is robust in the sense that a large degree of uncertainty can be tolerated in the components, as long as the contraction estimates are met for the network subsystems and their couplings. Computationally, one only needs to perform the maximizations involved in the computation of measures for individual (lower-dimensional) subsystems and couplings. In fact, applied recursively, one could even reduce the problem to a set $\log n$ computations for 2×2 matrices (for a system of dimension n). Among other features, contraction-based analysis has the advantage that no *a priori* knowledge is required of an attractor in order to perform stability analysis. Moreover, the approach can be also turned into *design* tool. For example, for set-point regulation or synchronization, once a system has been designed so that a particular state or subspace is flow-invariant, contraction ensures that all system trajectories converge to this desired point.

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